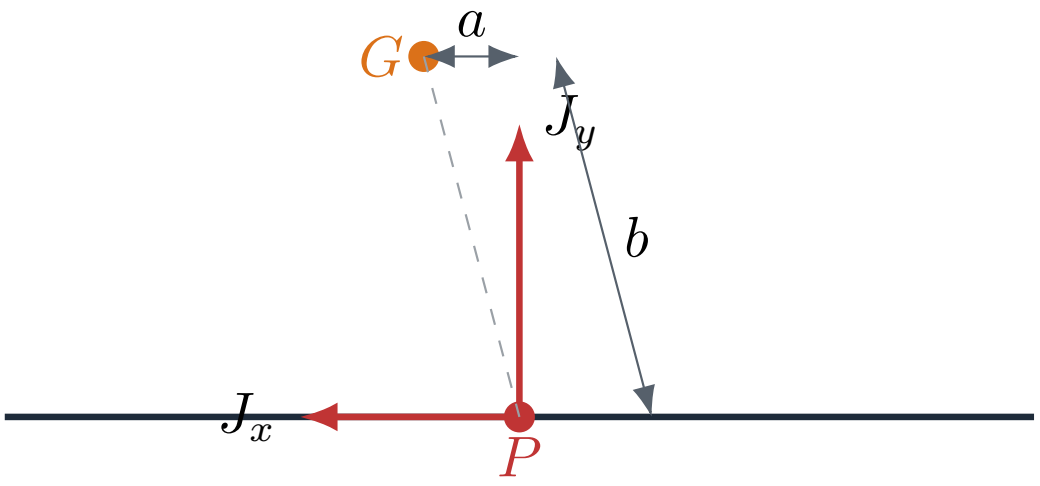

$$a = r_c \sin\theta, \quad b = r + r_c \cos\theta$$

$$v_{Py}^{+} = -v_{Py}^{-}$$

$$\mu_c = \frac{ab/r^2}{1 + b^2/r^2}$$